

SHIP DETECTION IN SAR IMAGES BASED ON AN IMPROVED FASTER R-CNN

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ABSTRACT

Deep learning has led to impressive performance on a variety of object detection tasks recently. But it is rarely applied in ship detection of SAR images. The paper aims to introduce the detector based on deep learning into this field. We analyze the advantages of the state-of-the-art Faster R-CNN detector in computer vision and limitations in our specific domain. Given this analysis, we proposed a new dataset and four strategies to improve the standard Faster R-CNN algorithm. The dataset contains ships in various environments, such as image resolution, ship size, sea condition, and sensor type, it can be a benchmark for researchers to evaluate their algorithms. The strategies include feature fusion, transfer learning, hard negative mining, and other implementation details. We conducted some comparison and ablation experiments on our dataset. The result shows that our proposed method obtains better accuracy and less test cost. We believe that SAR ship detection method based on deep learning must be the focus of future research.

Index Terms—Deep learning, SAR, ship detection, Faster R-CNN.

1. INTRODUCTION

Synthetic Aperture Radar (SAR) is an active radar that can provide high resolution images under all weather conditions. SAR images have been widely used for fishing vessel detection, ship traffic monitoring and immigration control. Numerous studies have been done to detect ships in SAR images^[1].

The Constant False-Alarm Rate (CFAR) and Viola & Jones are two common algorithms in this field. CFAR is widely used by setting a threshold so that we can find targets that are statistically significant above the background pixel while maintaining a constant false alarm rate. A function that fits distribution of clutter is first computed for determining the threshold. All pixels with their values higher than the threshold are defined as ship targets^[2]. After the pre-screening by CFAR, a discriminator is needed to reject the background. Viola and Jones is a seminal work which has a significant impact to object detection^[3]. It has three stages that makes it very fast: the integral image, the AdaBoost

classifier and the cascade structure. Since then, the feature extractor such as Histograms of oriented Gradients (HoG), Scale Invariant Feature Transform (SIFT), Speeded Up Robust Features (SURF), Local Binary Patterns (LBPs) are proposed and improve the performance further. Meanwhile the classifiers are also promoted rapidly, such as Boosting, Support Vector Machines (SVM) and their modifiers^[4-6]. The above two thoughts are effective in the past years. But in the era of deep learning, they face the problem of a low accuracy. So it is essential to adopt the detection method based on the deep learning in this domain.

Since AlexNet won ImageNet of image classification in 2012, neural network encounters another revival^[7]. Followed by ZF-Net, VGG-Net, GoogLeNet, and ResNet, the Convolutional Neural Network (CNN) continues to refresh the record of the classification task^[8]. Meanwhile as Ross and Kaiming put CNN into the detection task, they proposed a series of effective algorithms like Region based Convolutional Neural Network (R-CNN), Spatial Pyramid Pooling (SPP)-net, Fast R-CNN, and Faster R-CNN^[9-12]. The detection result becomes an unprecedented high level. Especially the Faster R-CNN, which has recently shown impressive results on various object detection benchmarks.

This paper presents the following contributions: In the beginning, we construct a dataset for ship detection in SAR images. We call it SAR Ship Detection Dataset (SSDD). SSDD contains ships in various environments. It can be a basic benchmark for researchers to evaluate their algorithms. Base on the dataset, we propose several improvements to boost the standard Faster R-CNN. Several comparisons and ablation experiments results show the efficiency of the Faster R-CNN and the effectiveness of our improvements.

2. RELATED WORK

Ross introduced a Region-based CNN (R-CNN) for object detection as shown in Fig.1. It has two stages: generating some object-agnostic proposals and training a regressor to refine the position of the bounding box. Approximately 2000 proposals pass through the CNN for extracting features per image. This causes a large computation. In order to relieve this problem, SPP-net^[10] and Fast R-CNN^[11] are proposed in order. They feed into the whole input image once to extract features, and project the region proposals to the final feature map. The Fast R-CNN is a special case of the SPP-net, which uses a single spatial pyramid pooling layer, and thus allows end-to-end fine-tuning a pre-trained ImageNet model.

Fast R-CNN accelerates the detection network by the ROI-pooling layer. However the region proposal step is out of the network, which still make the processing time a bottleneck. To accelerate the speed of the detector further, Faster R-CNN was proposed by Shaoqing^[12]. It has two modules as shown in Fig. 1. The first module is a fully convolutional network for generating proposals, called the Regional Proposal Network (RPN). The second module is Fast R-CNN detector whose task is refining the proposal generated in RPN. The key to reducing the computational burden is to share the convolutional layers from RPN and Fast R-CNN. Now the image only passes through the CNN once.

The procedure and the performance of the three algorithms are shown in Fig. 1.

Region proposal (Selective search)	Region proposal (Selective search)	Region proposal (Selective search)
Feature extraction (Deep Net)	Feature extraction (Deep Net)	Feature extraction (Deep Net)
Classification (SVM)	Classification + Rectangle refine (Deep Net)	Classification + Rectangle refine (Deep Net)
66% mAP 47s/image	70% mAP 3s/image	73.2% mAP 0.2s/image
R-CNN	Fast R-CNN	Faster R-CNN

Fig. 1 The three region-based CNN detectors.

3. DATASET AND THE PROPOSED METHOD

3.1. Dataset

We construct a dataset of ships in SAR images called SSDD. SSDD can be a benchmark for researchers to evaluate their algorithms. Anyone can get the dataset after sending the applying e-mail. For each of the ship, we predict the bounding box with a confidence score. We follow a similar procedure as PASCAL VOC^[13] to construct the dataset, as it is widely used in visual object detection. We divide the dataset into three parts (training set, validation set and the test set) with the proportion of 7:2:1.

We find that the dataset for detection is easier to construct than the classification. Because the task of detection is translation-variant, while the classification is translation-invariant. We try to include all the conditions as shown in Table 1. For example different resolutions, sizes, sea conditions, sensors and so on. This can make the detector more robust. But this also makes the detector very tough to get a high performance on the dataset.

TABLE 1 THE SSDD CONTAINS DIFFERENT KINDS OF SAR SHIP IMAGE.

sensors	polarization	scale	Ship
RadarSat-2 TerraSAR-X Sentinel-1	HH, VV, VH, HV	1:1 1:2 2:1	Different size and material
Sea condition	resolution	position	
Good and bad condition	1m-15m	in the sea and offshore	

Statistics for the number of ships and images in SSDD are given in Table 2. NoS is the abbreviation of number of ships, NoI is the abbreviation of number of images.

TABLE 2 THE CORRESPONDING RELATIONSHIPS BETWEEN NoS AND NoI IN SSDD.

NoS	1	2	3	4	5	6	7
NoI	725	183	89	47	45	16	15
NoS	8	9	10	11	12	13	14

NoI	8	4	11	5	3	3	0
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In SSDD, there are totally 1160 images and 2456 ships. The average number of ships per image are 2.12. The dataset would be expanded according the demands of the algorithms later. Compared to 9000+ images in PASCAL VOC with 20 categories, SSDD is big enough to train the one class detection model combined with many tricks to prevent over-fitting.

Fig. 2 shows the diversity of the ship in SAR images: (a) is a ship near the dock with 1m resolution, (b) is seven ships in the open sea, (c) is several warships near the dock, (d) is two ships near shore, (e) is eight ships in the open sea with 15m resolution.

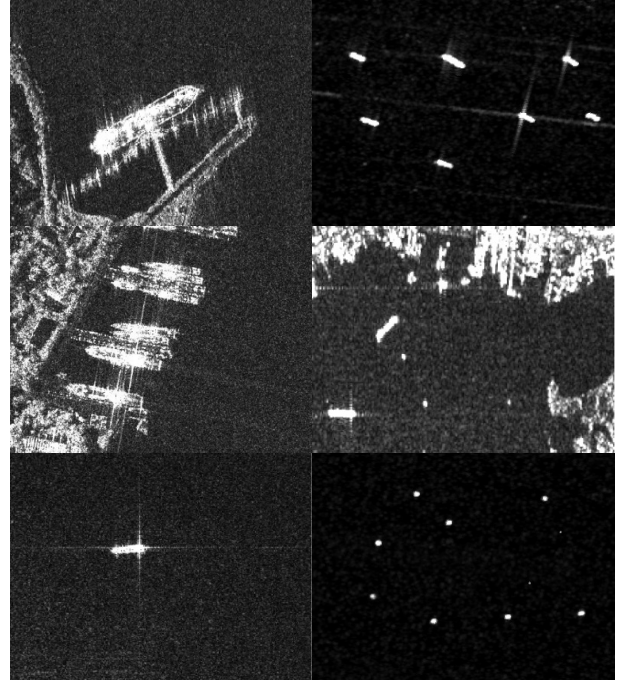


Fig. 2 Samples from SSDD.

As some small ships only have very few pixels in low resolution, sometimes it is hard to decide whether it is a ship or not. So if the number of pixel is more than three, we would regard it as a ship and make the annotation. We use the software called “labelimg”, which is convenience to annotate the ships with (x, y, h, w). In which (x, y) is the coordination of the rectangle’s box center point, h is the height of the rectangle, w is the width of the rectangle.

3.2. The Faster R-CNN and Its Limitations

In this section, we will briefly introduce the advantages and limitations of the Faster R-CNN. Faster R-CNN gets a good result on the common detection dataset. The RPN is its key module. RPN has a 3×3 fully convolutional layer connected to the end of the pre-trained meta-architecture (ZF, VGG16). It uses anchors with different scales and ratios to achieve translation invariance. An anchor is at each sliding location of the convolutional maps and thus at the center of each spatial window. The original paper has k = 9 anchors at each location, which includes 3 scales and 3 aspect ratios. If the

size of the convolutional feature map is $W \times H$, the number of the proposals are $W \times H \times k$. As RPN shares the convolutional layer with Fast R-CNN, the Faster R-CNN can finish the whole task in 0.2 second with the VGG-16.

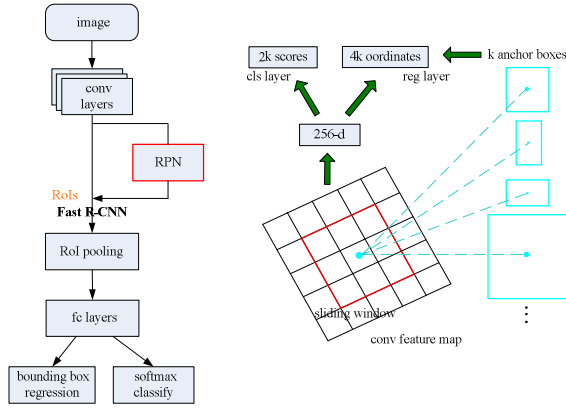


Fig. 3 Faster R-CNN and RPN (a) The flowchart of Faster R-CNN and (b) anchors of RPN.¹²

Faster R-CNN has a frame rate of 5fps on a GPU and gets the state-of-the-art object detection accuracy on PASCAL VOC 2007 (73.2% mAP) and 2012 (70.4% mAP) for the VGG-16 model. The success of the Faster R-CNN shows the potential of deep learning in the domain of object detection. Though Faster R-CNN gets a high accuracy on several multi-category dataset, it has not presented ideal results on SAR ship detection. We think that the unsatisfactory performance is due to the following reasons: Firstly, Faster R-CNN achieves state-of-the-art performance on PASCAL VOC datasets. They can detect objects such as persons, horses, or cars. These objects usually occupy the majority of an image. However, in our problem we are interested in detecting ships, which are usually small and low resolution as shown in Fig. 2. The detection network in Faster R-CNN has trouble to detect such small objects. The reason is the ROI-pooling layer builds features only from the last feature map. Thus the detector will have much difficulty to predict the object class and bounding box location. Secondly, as SAR image is quite different from the common optical image. The features learn from ImageNet may differ from SAR images. So transferring the whole pre-trained layers to SAR ship detection would not get a good performance. But the former layers that have lower feature may be common. Thirdly, because of the difference of the two dataset, many details should change to adopt the specific domain. For example, the anchor number, the proposal number, the dropout ratio and so on.

Based on the advantages and limitations of Faster R-CNN, we propose an improved method in Section 3.3.

3.3. The Proposed approach

The overview of our proposed method is shown in Fig. 4. We adopt the ZF-Net pre-trained on ImageNet, and fine-tune the model's last two convolutional layers using SSDD dataset. We fuse the different level features to improve the

ability to detect different size ships. Transfer learning, hard negative mining and other tricks are used to boost the performance further.

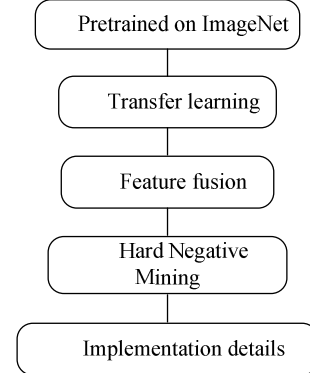


Fig. 4 Flowchart of the improved Faster R-CNN.

3.3.1. Feature Fusion

The sizes of ships in SSDD are usually changeable. It is hard for the Faster R-CNN to detect ships with different sizes, especially the small ones. This is because the receptive field in the latter layer is very large. This may omit some important features, resulting in a low accuracy^[14].

The detailed architecture of this approach is shown in Fig. 5. A fusion of low and high features can robustly detect different size of ships. We fuse the feature maps from convolutional layer 3 to layer 5 (Fig. 5).

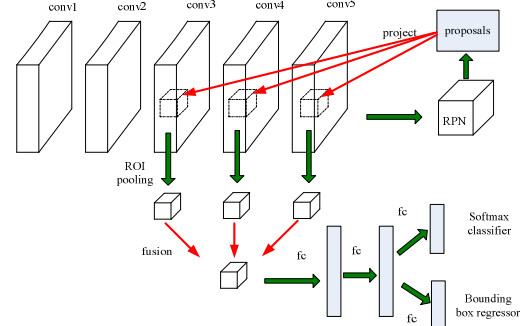


Fig. 5 Proposed feature fusion framework.

In the standard Faster R-CNN, the five convolutional layers would be followed by one ReLU, LRN and Max-pooling layer. But in our architecture, the last three convolutional layers only have one ReLU layer. The ROI pooling layer performs the function of Max-pooling.

The fusion includes the normalization and 1×1 convolution. Normalizing each ROI pooling tensor can reduce the scale differences between the following layers. It can prevent the 'larger' features dominating the 'smaller' ones and make the algorithm more robust. This modification stabilizes the system and increases the accuracy as illustrated in Section 4.3.

3.3.2. Transfer Learning

Many deep neural networks trained on natural images exhibit a phenomenon in common. On the first layer they learn features similar to Gabor filters and such features appear not to be specific to a particular dataset or task.

Features will eventually be translated from general to special by the last layers of the network^[15].

In fact, very few people can train a convolutional network from scratch. Instead, researchers usually use a model pre-trained on ImageNet. And regard the model as an initialization or a feature extractor. As the SSDD dataset is small and also different from ImageNet. We can't merely adopt the pre-trained model to our task. The best solution is to train the SVM classifier from activations somewhere earlier in the network in this scenarios.

ZF-Net is a variant of Krizhevsky's AlexNet. It reduces the first filter size from 11×11 to 7×7 , and changes the stride to 2. This net could maintain most of the information in the first and second layer. ZF-Net has five convolutional layers and three fc layers. Based on the analysis above, we fix the former three layers and fine-tune the latter two layers on SSDD. The ZF-Net architecture we use in our approach is shown in Fig. 6.

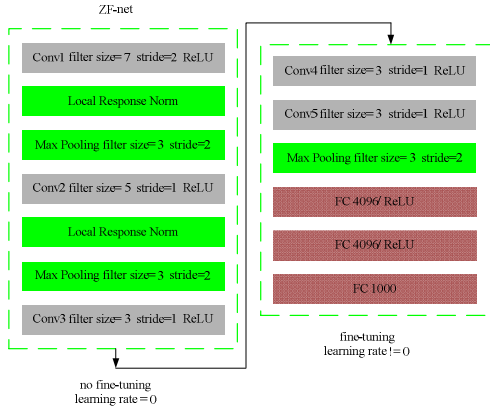


Fig. 6 ZF-Net architecture used in our approach.

3.3.3. Hard negative mining

Hard negatives are ships that are prone to be detected falsely by the detector. This strategy is an effective method to boost the performance of detection. What we need to do is to collect the hard negatives and feed them into the network again. In SSDD, we regard the ships with scores of 0.6-0.8 as the hard negatives. During the training, we add the hard negative target into the ROIs to fine-tune the model further. The boosted result will be shown in the following ablation experiments.

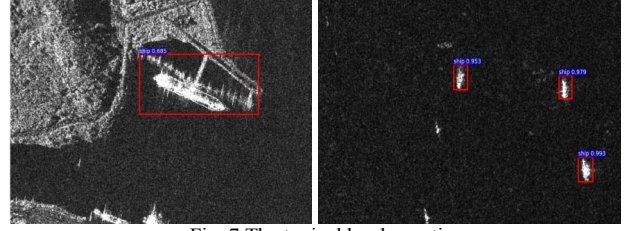
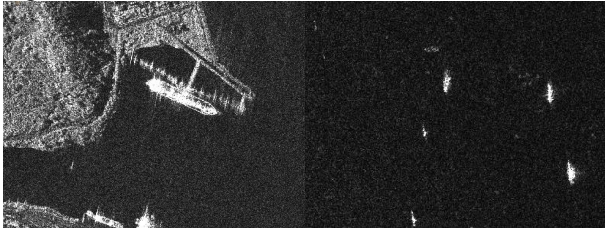


Fig. 7 The typical hard negatives.

Fig.7 shows several typical hard negatives. The left is the 757th sample in SSDD with a high resolution and big size. The right is the 59th sample in SSDD with a low resolution and small size.

3.3.4. Implementation details

In order to prevent over-fitting, we change the dropout from 0.5 to 0.6 and augment the data by flipping the positive ROIs during training. For anchors, we use 2 scales with box areas of 128^2 , 256^2 pixels, and 3 aspect ratios of 1:1, 1:2, and 2:1. As ships in SAR images are sparse, so we reduce the proposals from 300 to 50. This will speed up the method and without a big loss in accuracy. Other hyper-parameters of RPN are as in[12], and we adopt the publicly available code of to fine-tune the RPN. With the fine-tuned RPN, we adopt NMS with a threshold of 0.7 to filter the proposal regions.

4. EXPERIMENTS RESULT

4.1. Setup

We train the improved Faster R-CNN ship detection model on SSDD. For implementation, we adopt the Caffe^[16] framework to train our deep learning models. ZF-Net was selected to be our backbone CNN network, which had been pre-trained on ImageNet. The proposed method is evaluated on a 64 bits Ubuntu 14.04 computer with CPU Intel(R) Core(TM) i7-6770K @ 4.00GHz $\times$ 8 and NVIDIA GTX1080 GPU with 8G memory CUDA8.0 cuDNN5.0.

4.2. Comparison

We compare the proposed method with the standard Faster R-CNN. The metrics are Average Precision (AP) and average processing time per image.

From Table 3, we can see that the standard Faster R-CNN has an obvious advantage in both accuracy and time. The proposed method improves the AP from 70.1% to 78.8% with less time consuming.

TABLE 3. THE PERFORMANCE OF THE FOUR METHODS

methods	AP	time (ms)
Faster R-CNN	70.1%	198
Proposed	78.8%	173

We randomly choose some results of ship detection examples for different cases, as shown in Fig. 8.

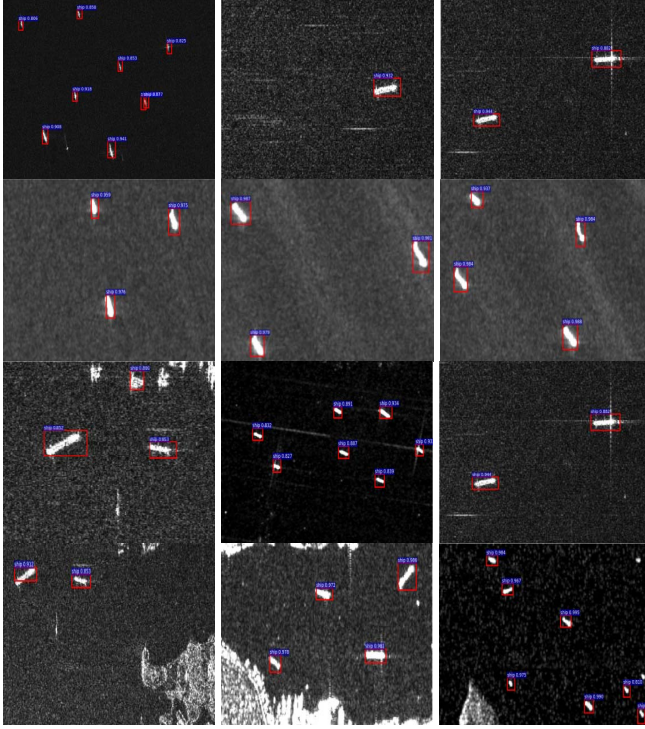


Fig. 8 Detection samples on the SSDD dataset.

4.3. Ablation experiment

In order to evaluate the proposed method further, we make some ablation experiments in Table 4. Our purpose is to examine the contributions of different strategies proposed in Section 3.3.

TABLE 4 RESULTS OF THE ABLATION EXPERIMENTS

methods	AP	time (ms) per image
Standard	70.1%	198
Improved	78.8%	183
Standard + Feature fusion	76.4%	213
Standard + Transfer learning	74.3%	203
Standard + Hard negative mining	75.6%	199
Standard + details	Dropout	71.6%
	NMS	69.1%
	Region proposal number	68.9%
	Dropout+NMS+Region proposal	68.6%

We examine the impact of feature fusion in 4th row of Table 4. By the fusion strategy illustrated in Section 3.3.1, the model can detect ships with different sizes. The average precision boost from 70.1% to 76.4%, while the test time increase less (15ms). Further we evaluate the performance of the transfer learning strategy, as shown in 5th row of Table 4. Since SSDD is a small dataset which has a big difference with the common object detection dataset. If we transfer the whole convolutional layers to our domain, the AP is about 70.1%. If we transfer the former three layers to our domain, and fine-tune the latter layer on SSDD. The average precision increase to 74.3% and the test time increase very less. The same as the hard negative mining. The average precision increases from 70.1% to 75.6%, but the test time nearly the same as the standard. This is because the strategy

improves the performance heavily by adding a lot of useful samples. This only increase the training time, and the test time nearly unchanged.

We examine the impact of anchor and region proposal number further. Instead of using the default setting (9 anchors) by standard faster RCNN, we compare this with our modification by deleting a size of 512×512 . This can increase the test time and the accuracy drop a little. The NMS increases the average precision a little and the test time nearly the same as standard Faster R-CNN. We change the dropout ratio from 0.5 to 0.6, the average precision increase by 1.5%. But when the ratio is 0.7 the performance descends a lot. This means the tricks we take prevent the over-fitting.

Finally, combining all the strategies above, the method achieves the best detection performance as shown in 3th row in Table 4.

5. CONCLUSION

We construct a dataset for ship detection of SAR images called SSDD. The SSDD is so far the first SAR images on public for researchers to evaluate the performance of their detectors. We also present an improved Faster R-CNN method to detect ships in SAR images. The proposed method adopts the standard Faster R-CNN as the meta-architecture. And change in four aspects according to SSDD. That is feature fusion, transfer learning, hard negative mining and some implementation details. The experiments conducted on SSDD demonstrate our proposed method has a better accuracy and is less time consuming.

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